

Agromorphological diversity and promising accessions of common bean (*Phaseolus vulgaris*) in the northeastern Peruvian Andes

José L. Huanambal-Gálvez^{1,2,3}, Wagner Meza-Maicelo^{1,2,3}, Victor H. Baldera-Chapoñan^{1,2,3}, Manuel Oliva-Cruz^{1,3}, Jorge L. Maicelo-Quintana⁴, and Flavio Lozano-Isla^{1,2,3,*}

¹Facultad de Ingeniería y Ciencias Agrarias, Universidad Nacional Toribio Rodríguez de Mendoza de Amazonas (UNTRM-A), Chachapoyas, Amazonas, Perú.

²Centro de Investigación e Innovación en Granos y Semillas, Universidad Nacional Toribio Rodríguez de Mendoza de Amazonas (UNTRM-A), Chachapoyas, Amazonas, Perú.

³Instituto de Investigación para el Desarrollo Sustentable de Ceja de Selva (INDES CES), Facultad de Ingeniería y Ciencias Agrarias (FICA), Universidad Nacional Toribio Rodríguez de Mendoza de Amazonas (UNTRM-A), Chachapoyas, Amazonas, Perú.

⁴Facultad de Ingeniería Zootecnista, Agronegocios y Biotecnología, Universidad Nacional Toribio Rodríguez de Mendoza de Amazonas (UNTRM-A), Chachapoyas, Amazonas, Peru

*Corresponding author: Flavio Lozano-Isla <flavio.lozano@untrm.edu.pe>



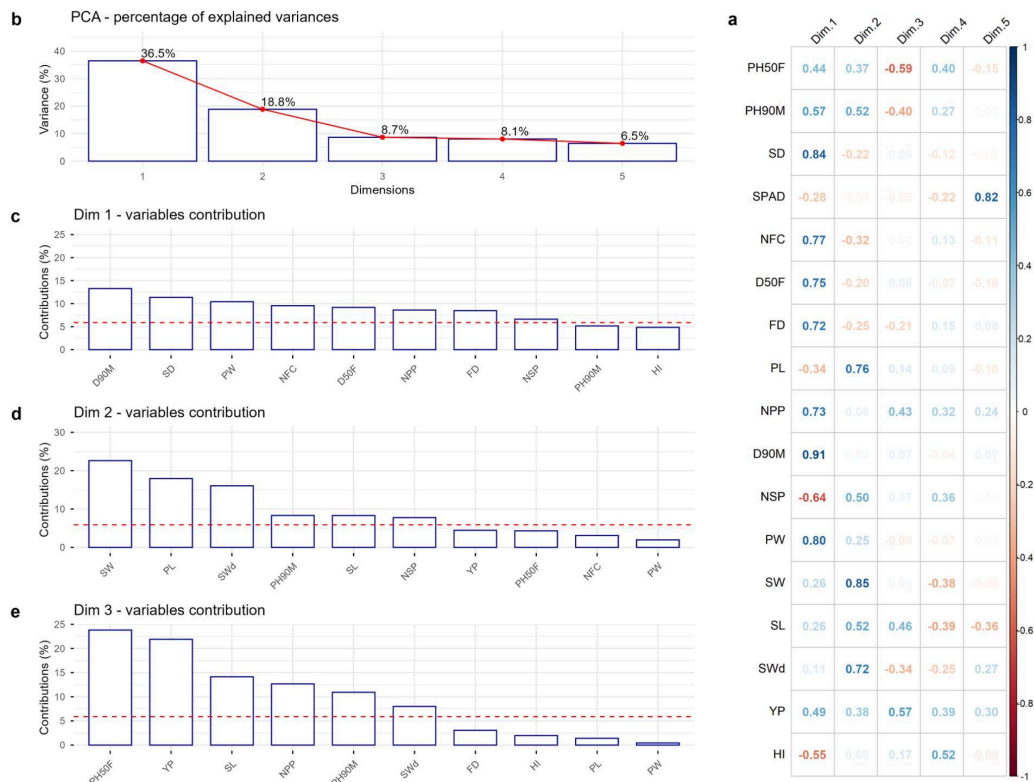
Supplementary Figure 1: Seed morphological diversity observed among the common bean (*Phaseolus vulgaris*) accession.

type ■ check ■ test ■ empty

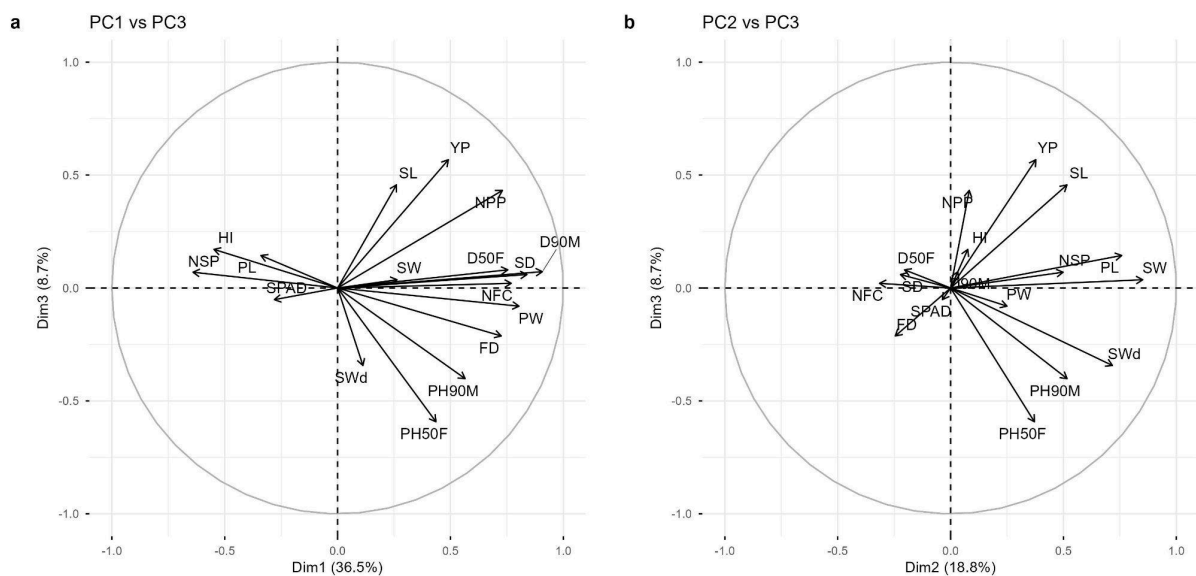
1	1001 UNTRM-335-1012	1002 UNTRM-335-1085	1003 UNTRM-335-1023	1004 UNTRM-335-1006	1005 UNTRM-335-1095	1006 UNTRM-335-1079
2	2001 UNTRM-335-1095	2002 UNTRM-335-1010	2003 UNTRM-335-1023	2004 UNTRM-335-1037	2005 UNTRM-335-1017	2006 UNTRM-335-1074
3	3001 UNTRM-335-1014	3002 UNTRM-335-1008	3003 UNTRM-335-1003	3004 UNTRM-335-1095	3005 UNTRM-335-1067	3006 UNTRM-335-1023
4	4001 UNTRM-335-1052	4002 UNTRM-335-1095	4003 UNTRM-335-1080	4004 UNTRM-335-1023	4005 UNTRM-335-1048	4006 UNTRM-335-1025
5	5001 UNTRM-335-1095	5002 UNTRM-335-1056	5003 UNTRM-335-1061	5004 UNTRM-335-1086	5005 UNTRM-335-1063	5006 UNTRM-335-1023
6	6001 UNTRM-335-1078	6002 UNTRM-335-1051	6003 UNTRM-335-1053	6004 UNTRM-335-1095	6005 UNTRM-335-1030	6006 UNTRM-335-1023
7	7001 UNTRM-335-1070	7002 UNTRM-335-1095	7003 UNTRM-335-1059	7004 UNTRM-335-1044	7005 UNTRM-335-1023	7006 UNTRM-335-1062
8	8001 UNTRM-335-1023	8002 UNTRM-335-1029	8003 UNTRM-335-1057	8004 UNTRM-335-1005	8005 UNTRM-335-1095	8006 UNTRM-335-1028
9	9001 UNTRM-335-1095	9002 UNTRM-335-1087	9003 UNTRM-335-1019	9004 UNTRM-335-1023	9005 UNTRM-335-1040	9006 UNTRM-335-1091
10	10001 UNTRM-335-1023	10002 UNTRM-335-1039	10003 UNTRM-335-1022	10004 UNTRM-335-1045	10005 UNTRM-335-1026	10006 UNTRM-335-1095
11	11001 UNTRM-335-1095	11002 UNTRM-335-1002	11003 UNTRM-335-1042	11004 UNTRM-335-1023	11005 UNTRM-335-1065	11006 UNTRM-335-1094
12	12001 UNTRM-335-1023	12002 UNTRM-335-1001	12003 UNTRM-335-1083	12004 UNTRM-335-1009	12005 UNTRM-335-1095	12006 UNTRM-335-1041
13	13001 UNTRM-335-1027	13002 UNTRM-335-1023	13003 UNTRM-335-1075	13004 UNTRM-335-1095	13005 UNTRM-335-1031	13006 UNTRM-335-1035
14	14001 UNTRM-335-1046	14002 UNTRM-335-1066	14003 UNTRM-335-1023	14004 UNTRM-335-1032	14005 UNTRM-335-1095	14006 UNTRM-335-1033
15	15001 UNTRM-335-1038	15002 UNTRM-335-1095	15003 UNTRM-335-1034	15004 UNTRM-335-1058	15005 UNTRM-335-1023	15006 UNTRM-335-1060
16	16001 UNTRM-335-1016	16002 UNTRM-335-1095	16003 UNTRM-335-1015	16004 UNTRM-335-1023	16005 UNTRM-335-1088	16006 UNTRM-335-1013
17	17001 UNTRM-335-1095	17002 UNTRM-335-1076	17003 UNTRM-335-1071	17004 UNTRM-335-1007	17005 UNTRM-335-1020	17006 UNTRM-335-1023
18	18001 UNTRM-335-1047	18002 UNTRM-335-1023	18003 UNTRM-335-1021	18004 UNTRM-335-1095	18005 UNTRM-335-1054	18006 UNTRM-335-1084
19	19001 UNTRM-335-1095	19002 UNTRM-335-1064	19003 UNTRM-335-1023	19004 UNTRM-335-1049	19005 UNTRM-335-1090	19006 UNTRM-335-1050
20	20001 UNTRM-335-1095	20002 UNTRM-335-1092	20003 UNTRM-335-1011	20004 UNTRM-335-1089	20005 UNTRM-335-1024	20006 UNTRM-335-1023
21	21001 UNTRM-335-1068	21002 UNTRM-335-1081	21003 UNTRM-335-1093	21004 UNTRM-335-1095	21005 UNTRM-335-1004	21006 UNTRM-335-1023
22	22001 UNTRM-335-1095	22002 UNTRM-335-1072	22003 UNTRM-335-1023	22004 UNTRM-335-1073	22005	22006 UNTRM-335-1018
23	23001 UNTRM-335-1043	23002 UNTRM-335-1077	23003 UNTRM-335-1095	23004	23005 UNTRM-335-1023	23006 UNTRM-335-1036
24	24001 UNTRM-335-1069	24002 UNTRM-335-1055	24003	24004 UNTRM-335-1023	24005 UNTRM-335-1082	24006 UNTRM-335-1095
	1	2	3	4	5	6

Columns

Supplementary Figure 2: Field layout of the Augmented Design established for the agronomic and morphological characterization of the common bean (*Phaseolus vulgaris*) genotypes. The experimental matrix consisted of 26 rows corresponding to blocks and 6 columns representing the accessions within each block.



Supplementary Figure 3: Principal Component Analysis (PCA) of the common bean (*Phaseolus vulgaris*) accessions based on the 17 evaluated variables. (a) Percentage of explained variance. (b) Variable contributions to Dimension 1. (c) Variable contributions to Dimension 2. (d) Correlation matrix between variables and dimensions.



Supplementary Figure 4: Principal Component Analysis (PCA) showing the correlations among the 17 evaluated variables across the third dimension (Dim3). The figure presents the relationships between Dim1 and Dim3 and between Dim2 and Dim3.